Python Assignment
Topic-4: Conditional Statements

Q1. Introduction to conditional statements: if, else, elif.

Ans.

Conditional statements are used to make decisions in a program. They allow the program to execute different blocks of code based on conditions. The 'if' statement is used to check a condition. If the condition is true, the code inside it runs. The 'else' statement is used when the condition is false. The 'elif' statement is used to check multiple conditions. These statements help in controlling the flow of the program. They are very important for making logical decisions in Python programs.

Example:

```
x = 10
if x > 5:
   print("Greater")
else:
   print("Smaller")
```

Q2. Nested if-else conditions.

Ans.

Nested if-else means using one if-else statement inside another if-else statement. It is used when we have multiple conditions to check. It helps in solving complex decision-making problems. In nested if, the second condition is checked only if the first condition is true. This makes the program more flexible. Nested conditions should be used carefully to avoid confusion. Proper indentation is very important while using nested if-else. It helps in writing structured and readable code.

Example:

```
x = 10
if x > 5:
   if x < 20:
      print("Between 5 and 20")
```